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Second-Hand Product Escrow Marketplace

PROBLEM STATEMENT 39

RE-DELIVERABLES

1. Problem Statement Analysis

Overview: The system is a secure peer-to-peer (P2P) marketplace for second-hand goods in which the buyer's payment is held in escrow rather than transferred directly to the seller. An independent third-party inspection agent physically verifies the condition of the item against the listing, and the escrowed funds are released to the seller only after the buyer signs off on the inspection outcome. This structure removes the trust gap that normally exists between strangers trading used goods online.

Key Stakeholders / Actors

- **Buyer** – browses listings, funds escrow, reviews the inspection report, and approves/rejects the item.
- **Seller** – lists the second-hand product and receives payout once escrow is released.
- **Inspection Agent** – independently verifies the physical condition of the product and submits a certified sign-off report.
- **Payment Gateway** (external system) – holds and transfers escrow funds on the platform's instruction.
- **Admin** – monitors disputes and oversees escrow/dispute resolution workflow.

Core Problem Being Solved

- Buyers have no way to verify a used product's real condition before paying in full.
- Sellers risk non-payment or chargebacks after shipping/handing over goods.
- There is no neutral, evidence-backed checkpoint between payment and fund release.
- Disputes over item condition currently have no structured, auditable resolution path.

Key Constraints Identified

- Funds must remain locked in escrow until dual verification (inspector + buyer) is complete.
- Inspection outcomes must be evidence-based (not just a text status) to be trustworthy.
- Disputes must freeze fund movement immediately to prevent premature payout.
- All financial and escrow data must be secured to a high standard given its sensitivity.

2. Requirement List (5 FR + 2 NFR)

Functional and nonfunctional requirements were elicited directly from the problem statement above. FR-001 and NFR-001 were provided; FR-002 to FR-005 and NFR-002 were derived to cover the full escrow → inspection → release / dispute lifecycle.

Functional Requirements

- FR-001: The system shall lock buyer payment in escrow and transition the order state to 'Under Inspection' until an inspector submits a physical verification sign-off.
- FR-002: The system shall allow the inspection agent to submit a certification report (pass/fail) with photographic evidence attached for each item under inspection.
- FR-003: The system shall allow the buyer to approve or reject the item within 48 hours of receiving the inspection report.
- FR-004: The system shall release escrowed funds to the seller once the buyer approves the item, or automatically after 48 hours if the inspection passed and the buyer has not responded.
- FR-005: The system shall allow the buyer to raise a dispute if the inspection report or item condition is contested, routing the order to a dispute-resolution queue.

Nonfunctional Requirements

- NFR-001: Dispute resolution tickets must freeze escrow funds immediately and prevent account withdrawals during active investigation. (Performance & Security)
- NFR-002: The system shall encrypt all payment and escrow transaction data both in transit and at rest using industry-standard encryption (e.g., TLS 1.2+ / AES-256). (Security)

3. Requirements Table

Req ID	Type	Description	Priority	Acceptance Criteria	Rationale
FR-001	Functional	The system shall lock buyer payment in escrow and transition order state to 'Under Inspection' until an inspector submits physical verification sign-off.	High	Pass: funds held securely and released only on dual verification. Fail: funds released prematurely to seller.	Core trust mechanism of the escrow marketplace.
FR-002	Functional	The system shall allow the inspection agent to submit a certification report (pass/fail) with	High	Pass: report cannot be submitted without at least 1 photo and a pass/fail value. Fail: report	Ensures inspection outcomes are auditable and evidence-based.

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		photographic evidence for each item under inspection.		accepted with no evidence attached.	
FR-003	Functional	The system shall allow the buyer to approve or reject the item within 48 hours of receiving the inspection report.	High	Pass: buyer decision is captured and order state updates within the 48-hour window. Fail: decision not recorded or window not enforced.	Preserves buyer's final say over release of their funds.
FR-004	Functional	The system shall release escrowed funds	Medium	Pass: payout is initiated within 24 hours of the	Automates settlement while

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		to the seller upon buyer approval, or automatically after 48 hours if inspection passed and the buyer did not respond.		approval/auto-release trigger. Fail: funds remain locked beyond 24 hours after the trigger.	protecting the buyer's response window.
FR-005	Functional	The system shall allow the buyer to raise a dispute if the inspection report or item condition is contested, routing the order to a dispute-	Medium	Pass: a dispute ticket is created with the order ID, escrow release is frozen, and it appears in the admin queue within 1 minute. Fail: no ticket created or	Provides a recourse path that preserves marketplace trust.

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		resolution queue.		release continues.	
NFR-001	Nonfunctional	Dispute resolution tickets must freeze escrow funds immediately and prevent account withdrawals during active investigation.	High	Pass: benchmarking under simulated peak load confirms funds freeze within target latency with no withdrawal possible during investigation.	Prevents financial loss while a dispute is being investigated.
NFR-002	Nonfunctional	The system shall encrypt all payment and escrow transaction data both in	High	Pass: security audit confirms TLS 1.2+ in transit and AES-256 at rest, with no plaintext	Protects sensitive financial data and builds user trust in

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		transit and at rest using industry-standard encryption (e.g., TLS 1.2+ / AES-256).		financial data found. Fail: any unencrypted transmission or storage detected.	the escrow process.

4. Requirements Traceability Matrix (RTM)

The RTM traces each requirement back to its source need in the problem statement and forward to the use case and verification method that will confirm it has been satisfied.

Req ID	Source / Business Need	Related Use Case	Verification Method	Status
FR-001	Buyer payments must not go directly to seller before verification.	UC-01: Fund Order into Escrow	System test: order state + fund lock check	Elicited
FR-002	Buyer needs proof of product condition before deciding to accept.	UC-02: Submit Inspection Report	Functional test: report requires evidence to submit	Elicited

Req ID	Source / Business Need	Related Use Case	Verification Method	Status
FR-003	Buyer must retain final control over releasing their money.	UC-03: Review & Approve/Reject Item	Functional/UAT: 48-hour decision window test	Elicited
FR-004	Seller must be paid promptly once verification/approval is done.	UC-04: Release Escrow Funds	Integration test with Payment Gateway on release timing	Elicited
FR-005	Buyer needs recourse if the item or report is disputed.	UC-05: Raise Dispute	Functional test: ticket creation + fund freeze trigger	Elicited
NFR-001	Funds must be protected the moment a dispute is opened.	UC-05: Raise Dispute (extends)	Load test: freeze latency under simulated peak load	Elicited
NFR-002	Financial/escrow data is highly sensitive and must be secured.	Cross-cutting: all escrow & payment use cases	Security audit / penetration test	Elicited